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COMPUTER SCIENCE PROGRAMME

Master's Thesis

VIRTUAL HISTORY TEACHER: PEDAGOGICAL CHATBOT AND AUTOMATIC DETECTION OF CONTRADICTIONS BETWEEN HISTORICAL SOURCES

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Rezumat

Lucrarea prezintă proiectarea, implementarea și evaluarea unui sistem conversațional destinat sprijinirii predării istoriei la nivelul clasei a VII-a din învățământul românesc. Sistemul pornește de la o singură intrare furnizată de utilizator — o întrebare — pe care o tratează prin două module complementare. Primul modul este un asistent de tip *Retrieval-Augmented Generation* (RAG) care răspunde la întrebări folosind exclusiv un corpus de lecții indexat. Acesta cuprinde un pipeline de ingerare a datelor (segmentare semantică, extragere de metadate și recunoaștere de entități, sumarizare ierarhică, vectorizare), un subsistem de înțelegere a interogării (clasificare, verificarea acoperirii și extragerea filtrelor) și un mecanism de regăsire hibridă (filtrare pe metadate, căutare vectorială și reordonare cu un *cross-encoder*), urmat de generarea unui răspuns ancorat în surse, cu citări. Al doilea modul abordează problema contradicțiilor dintre surse istorice, unde subiectivitatea și interpretarea pot introduce inconsistențe. Acesta aplică un set de întrebări canonice asupra mai multor surse și detectează contradicțiile prin combinarea unei verificări structurale (date și valori numerice), a inferenței în limbaj natural și a unui evaluator de tip model de limbaj. Sistemul a fost supus unei evaluări cantitative pe cele două module: pentru modulul de regăsire, analiza s-a realizat pe trei niveluri, în timp ce pentru modulul de detecție a contradicțiilor s-au utilizat metrici de regăsire, calculate pe un corpus adnotat manual. Rezultatele indică o regăsire precisă și răspunsuri fidele contextului, validând utilitatea abordării pentru un context educațional.

Abstract

This thesis presents the design, implementation, and evaluation of a conversational system intended to support the teaching of history at the seventh-grade level in the Romanian curriculum. The system is built to operate from a single user input — a question — which it processes through two complementary modules. The first module is a *Retrieval-Augmented Generation* (RAG) assistant that answers questions grounded exclusively in an indexed lesson corpus. It comprises a data-ingestion pipeline (semantic chunking, metadata and named-entity extraction, hierarchical summarization, embedding), a query-understanding subsystem (classification, coverage checking, and filter extraction), and a hybrid retrieval mechanism (metadata filtering, vector search, and cross-encoder reranking), followed by source-grounded answer generation with citations. The second module addresses contradictions between historical sources, where subjectivity and interpretation may introduce inconsistencies. It applies a set of canonical questions across multiple sources and detects contradictions by combining a structural check (dates and numerical values), natural-language inference, and a language-model judge. The system underwent quantitative evaluation across its two modules: the retrieval module was analyzed at three levels, while the contradiction detection module was assessed using retrieval metrics, computed against a manually annotated corpus. The results indicate accurate retrieval and context-faithful answers, supporting the usefulness of the approach in an educational setting.

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Chapter 1

Introduction

1.1 Type of thesis and subfield

The master’s thesis is situated within the field of artificial intelligence, focusing specifically on natural language processing (NLP) and information retrieval (IR). On the application side, it falls under conversational systems and artificial intelligence for education. The topic combines recent techniques based on large language models with classical information retrieval methods and with models that capture semantic relations between sentences, integrated into a single application with a teaching purpose.

1.2 Overview of the topic

Large language models have shown a remarkable ability to generate fluent and plausible text, but they have two fundamental limitations when used as sources of knowledge: a tendency to produce incorrect yet credible statements (a phenomenon known as *hallucination*), as well as an inability to access post-training information without an external retrieval mechanism. These limitations are particularly problematic in an educational setting, where factual accuracy is essential and the target audience — lower-secondary students — lacks the means to verify the claims it receives.

The *Retrieval-Augmented Generation* (RAG) paradigm answers these limitations by combining a generative model with a retrieval mechanism: before producing an answer, the system searches a document corpus for the relevant passages and supplies them to the model as *context*, conditioning generation on information from verified sources.

This thesis builds such a system, adapted to the teaching of history in the seventh grade. Starting from a single input — a question asked by a student or a teacher —, the system provides two functionalities, treated as separate modules (Figure 1.1). **Module 1** is a RAG assistant that generates an answer drawn from the indexed lessons, designed to match the comprehension skills of a seventh-grade student. **Module 2** tackles a

humanities-specific problem: contradictions between sources. Given several sources on the same topic, the module automatically determines whether the claims extracted from them contradict one another, providing a tool for the critical analysis of historical information.

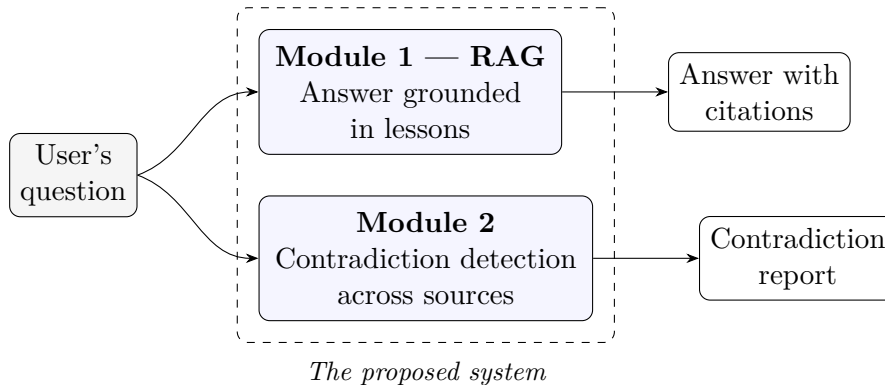


Figure 1.1: High-level architecture of the system. A single input (the question) is handled by two independent modules, each with its own output.

1.3 Aim and motivation

The aim of the thesis is to build a system that answers history questions with high accuracy and transparency, producing answers that can be traced back to their source and that can also flag divergences between sources. The motivation is driven by two primary factors. On the practical side, general-purpose conversational tools are not suited to a strict educational setting: they do not guarantee that the information comes from validated teaching materials, and they offer no means of verification. On the methodological side, and specific to artificial intelligence, history is a discipline in which interpretation, subjectivity, and *bias* are often present, so automatic detection of contradictions between sources is a relevant case study for the problem of verifying factual consistency across documents.

1.4 Original contributions

The main contributions of this thesis are the following:

- the design and implementation of a complete data-ingestion pipeline that turns unstructured history lesson text into a hierarchical, semantically enriched, embedded representation stored in a document-oriented database;
- the development of a three-stage query-understanding subsystem (question classification, corpus coverage checking, and filter extraction), which controls cost and avoids answering out-of-domain questions;

- the construction of a hybrid retrieval mechanism that combines metadata filtering with vector search and with *cross-encoder* reranking;
- the design of a contradiction-detection module that uses canonical questions to align sources and combines a structural check, natural language inference, and an LLM judge;
- a quantitative evaluation of both modules: three levels for retrieval, and metrics against a manually annotated corpus for contradiction detection.

1.5 Structure of the thesis

The thesis is organized into five chapters. Chapter 1 introduces the topic, the aim, and the contributions. Chapter 2 presents the theoretical background of the thesis and reviews related work. Chapter 3 describes Module 1, the RAG assistant, in detail, from data ingestion to answer generation and evaluation. Chapter 4 presents Module 2, dedicated to detecting contradictions between sources, together with its evaluation on an annotated corpus. Chapter 5 summarizes the results, discusses limitations, and proposes directions for future work.

Chapter 2

Preliminaries

This chapter presents the theoretical and technological background of the thesis, together with a review of related work. The aim is to provide the context needed to understand the design decisions described in Chapters 3 and 4.

2.1 Large language models and their limitations

Large language models (LLMs) are neural networks based on the *Transformer* architecture [13], trained on vast text corpora to predict the next word in a sequence. At scale, this apparently simple task leads to general linguistic and reasoning abilities. The knowledge of such a model, however, is stored implicitly in its parameters and has two drawbacks that matter for factual applications. First, the model can generate incorrect statements with the same fluency as correct ones — *hallucinations* — without signalling any uncertainty. Second, the model’s knowledge is fixed at training time and cannot be updated without either retraining or an external search mechanism (real-time access to web sources). In an educational setting it is also essential that information come from an agreed and verifiable source, a condition that a generative model cannot guarantee.

2.2 Vector representations and semantic similarity

An *embedding* is a representation of a piece of text as a vector of real numbers $\mathbf{v} \in \mathbb{R}^d$, such that texts with similar meaning are mapped to nearby vectors. Unlike representations based on word matching, embeddings capture *semantic* similarity: two different phrasings of the same idea can receive nearby vectors even if they share no word. Closeness between two vectors is usually measured by cosine similarity:

$$\text{sim}(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|} = \frac{\sum_{i=1}^d u_i v_i}{\sqrt{\sum_{i=1}^d u_i^2} \sqrt{\sum_{i=1}^d v_i^2}}. \quad (2.1)$$

The value lies between -1 and 1 ; the closer it is to 1 , the more similar the texts are. In this thesis, embeddings are produced by a dedicated model (`text-embedding-3-large`) [10] and form the basis of semantic search.

2.3 Retrieval-Augmented Generation

Retrieval-Augmented Generation [8] is a paradigm that combines a retrieval component with a generative model. Instead of relying solely on the model’s internal knowledge, the system first retrieves the passages relevant to the question from an external corpus, then passes them to the model as *context*, conditioning answer generation on those passages. The advantages are direct: answers become *grounded* in verifiable sources, they can carry citations, and the knowledge base can be updated by editing the corpus alone, without retraining the model. The quality of a RAG system depends critically on the quality of the retrieval component: if the passages supplied as context are irrelevant or incomplete, not even the most capable generative model can produce a correct answer.

2.4 Lexical retrieval, dense retrieval, and hybrid approaches

Information retrieval has two main families of methods. *Lexical* methods rely on matching terms between the query and the document. A classic representative is the Okapi BM25 scoring function [11], which, for a query Q made up of the terms $q_1, \dots, q_n$ and a document D , computes:

$$\text{BM25}(Q, D) = \sum_{i=1}^n \text{IDF}(q_i) \cdot \frac{f(q_i, D) (k_1 + 1)}{f(q_i, D) + k_1 \left(1 - b + b \frac{|D|}{\text{avgdl}} \right)}, \quad (2.2)$$

where $f(q_i, D)$ is the frequency of term q_i in the document, $|D|$ the document length, avgdl the average document length in the collection, and k_1 and b are calibration parameters. Lexical methods are fast and precise when terms coincide, but they do not capture synonyms or paraphrases.

Dense methods represent the query and the documents as embeddings and retrieve documents by vector similarity, following Equation (2.1). This approach, popularized by *Dense Passage Retrieval* [7], captures semantic similarity, but it can miss exact matches on rare entities or specific terms. *Hybrid* approaches combine the two families to benefit at once from lexical precision and semantic generalization; the system described in Chapter 3 adopts such a hybrid strategy, complemented by filtering on structured metadata.

2.5 Reranking with cross-encoders

Initial retrieval is designed for speed and typically returns a broader candidate set than needed. To select the most relevant passages, a *reranking* stage is applied. Unlike *bi-encoder* models, which encode the query and the document independently, a *cross-encoder* processes the (query, document) pair jointly and produces a more precise relevance score [9]. The higher computational cost is acceptable, because reranking is applied only to candidates already selected. In this thesis, the stage is handled by a dedicated reranking service.

2.6 Named entity recognition

Named entity recognition (NER) is the task of identifying and classifying instances of entities in text — people, organizations, places, dates, events. In a historical setting, entities (figures, battles, treaties, years) are natural anchors both for filtering retrieval and for comparing claims across sources. A difficulty specific to Romanian is the inflection of proper nouns (for example, “România”, “României”), which requires lemmatization (reduction to the base form) so that the same entity is recognized across different sources. The thesis uses a statistical NER model for Romanian from the spaCy library [6].

2.7 Natural language inference

Natural language inference (NLI) is the task of determining the relation between a *premise* and a *hypothesis*, classifying it into one of three categories: *entailment*, *neutral*, or *contradiction*. Modern NLI models are cross-encoders trained on large corpora such as MultiNLI [14] and, in the multilingual case, XNLI [2]. The ability to recognize the *contradiction* label makes NLI a tool directly applicable to detecting inconsistencies between sources. The model produces a probability distribution over the three labels by applying the *softmax* function to the output scores $\mathbf{z} = (z_1, z_2, z_3)$:

$$p_j = \frac{e^{z_j}}{\sum_{k=1}^3 e^{z_k}}. \quad (2.3)$$

Because the inference relation is directional (premise $\rightarrow$ hypothesis), each pair is evaluated in both directions in this thesis, keeping the maximum contradiction probability. The model used is a multilingual variant based on DeBERTaV3 [5], trained for NLI.

2.8 Evaluating retrieval and generation systems

Evaluating a RAG system targets both the retrieval and the generation component. For retrieval, classical information retrieval metrics are used. *Hit@k* measures the fraction of queries for which at least one relevant document appears in the top k results. *Mean Reciprocal Rank* (MRR) rewards placing the relevant document as high as possible in the ranking:

$$\text{MRR} = \frac{1}{|Q|} \sum_{i=1}^{|Q|} \frac{1}{\text{rank}_i}, \quad (2.4)$$

where rank_i is the position of the first relevant result for query i .

Evaluating the generative component is harder, since an answer can be correctly phrased in many ways. An increasingly common approach is *LLM-as-a-Judge*, in which a strong language model scores answers against defined criteria, such as faithfulness to the context and factual accuracy [15]. To avoid the self-preference bias characteristic of language models, the judge used in the evaluation (Claude Sonnet 4.6) was a different model from the one that generated the answers (gpt-4o-mini), removing an LLM’s tendency to overrate the quality of its own output. This methodology covers the generative component of the RAG system [3]. For the classification components (question routing, for example) and for contradiction detection, standard metrics are used: precision, recall, and the F_1 score:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \quad F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}, \quad (2.5)$$

where TP , FP , and FN are the numbers of correct detections, false positives, and false negatives, respectively.

2.9 Related work

RAG systems have become the standard for conversational applications grounded in a knowledge base, from documentation assistants to educational tutors. The educational direction emphasizes answer faithfulness and adapting the language to the student’s level, both addressed explicitly in this thesis through grounded generation with citations and through style instructions.

The closest commercial tool to Module 1 is NotebookLM [4], a RAG assistant that answers, with citations, solely from the sources a user uploads. The system proposed here shares this grounding property, but differs through choices specific to the educational setting: (i) an explicit and *measurable* domain-filtering mechanism, which rejects questions outside history or not covered by the corpus *before* retrieval (Section 3.3), unlike a generalist assistant that will try to answer from whatever material is available; (ii) *hybrid*

retrieval with structured metadata filters (years, entities, locations), which gives high precision on factual questions instead of a purely semantic retrieval; (iii) explicit control of the *reading level*, tuned to a seventh-grade student; and (iv) *an open pipeline that can be evaluated component by component* (Section 3.6), unlike a closed system that can be neither inspected nor adapted to a syllabus. At the system level, the thesis adds a second module, for detecting contradictions between sources, absent from such tools.

Factual consistency between sources is studied in the literature on automated fact-checking, where the classic task evaluates a claim against a corpus of evidence, as in the FEVER dataset [12]. Unlike that formulation, the case of historical sources has no reference source of official standing to settle the dispute between a true statement and a fabricated (false) one. Without an external arbiter, the problem becomes one of logical compatibility and reduces to deciding whether two statements can be true at the same time. The approach proposed in Chapter 4 handles this problem by aligning sources on shared questions and by combining a structural check, natural language inference, and an LLM judge, aiming to reduce the false positives commonly observed in purely semantic methods.

Chapter 3

Module 1: The RAG-based conversational assistant

This chapter describes the first module of the system: an assistant that answers history questions using only an indexed lesson corpus. The module follows the RAG paradigm and is organized in four stages: data ingestion, query understanding, retrieval, and answer generation. The chapter closes with a quantitative evaluation of the module.

3.1 Overall architecture and data flow

The system takes a natural language question as input and produces an answer grounded in the lessons, accompanied by citations. Before this flow can run, the lesson corpus must be prepared through an ingestion process that turns it into a structured, embedded representation. At runtime, the question passes through a query-understanding subsystem, which decides whether it is worth processing and extracts cues useful for retrieval; the retrieval mechanism then selects the relevant passages, and the generative model formulates the answer from them. The corpus covers the seventh-grade history syllabus, with lessons on contemporary history (for example, the economic and social transformations of the nineteenth century, the two World Wars, and the postwar world).

3.2 Data-ingestion pipeline

Answer quality depends on the quality of the corpus representation. The ingestion pipeline (Figure 3.1) turns the raw text of each lesson into a hierarchical, semantically enriched, embedded structure, stored in a document-oriented database.

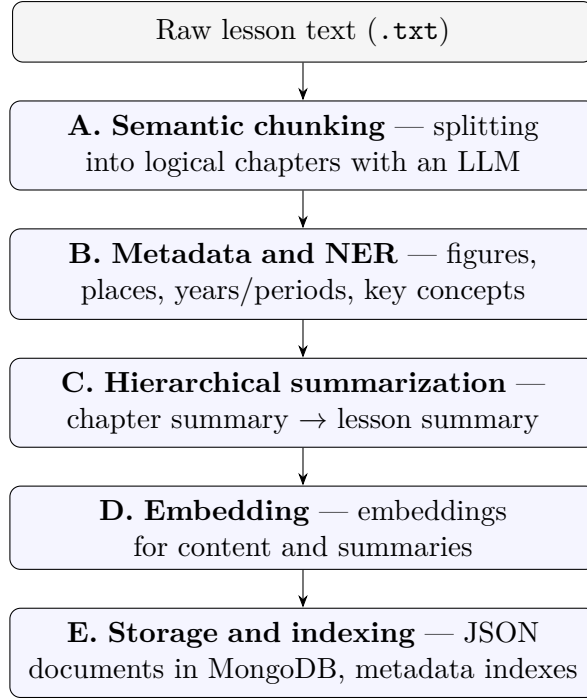


Figure 3.1: The data-ingestion pipeline. Each stage enriches the lesson representation.

3.2.1 Semantic chunking

The lesson text is split into logical units (chapters). When the text carries no explicit markers, a cheap language model (`gpt-4o-mini`) identifies the thematic transitions and delimits the chapters. Semantic chunking avoids arbitrary fragmentation (for example, on a fixed character count), which could cut an idea in half and hurt retrieval quality.

3.2.2 Metadata extraction and entity recognition

For each chapter we extract structured metadata: historical figures, locations, years and periods, and key concepts. This metadata is essential for filtering retrieval later, since it allows the search to be narrowed to the passages that mention a given entity, year, or place. The metadata is normalized (for example, special characters “ă”, “â”, “î”, “ș”, “ț” are stripped for robust matching and years are represented as intervals) so that it can be used efficiently in database queries.

3.2.3 Hierarchical summarization

Summarization happens at two levels. First, a concise summary is generated for each chapter; then, by concatenating the chapter summaries, a global summary of the lesson is produced (*master summary*). This hierarchical structure gives representations at different granularities: the lesson summary suits topic-level retrieval, while chapter content provides the detail needed for specific answers.

3.2.4 Embedding

For each chapter we compute embeddings with the `text-embedding-3-large` model: one for the chapter content and one for its summary. We deliberately avoid embedding the entire lesson text in a single vector, since such a representation would dilute meaning. Lesson-level retrieval instead relies on the summary vector. The vectors are stored together with the text and the metadata.

3.2.5 Storage and indexing

The final representation is saved in a MongoDB database, whose JSON document structure fits the lesson–chapter hierarchy and the flexible metadata schema naturally (Listing 3.1). Indexes are created on the metadata fields (years, year ranges, locations, entities, key concepts), which allow fast filtering at retrieval time.

Listing 3.1: Simplified structure of a lesson in MongoDB.

```
{
  "lesson_id": "lesson_02",
  "title": "Primul Razboi Mondial",
  "lesson_summary": "...",
  "chapters": [
    {
      "chapter_id": 1,
      "content": "Textul original al capitolului...",
      "summary": "Rezumatul capitolului...",
      "metadata": {
        "years": ["1914", "1918"],
        "entities": ["Franz Ferdinand", "Gavrilo Princip"],
        "locations": ["Sarajevo", "Europa"],
        "key_concepts": ["asasinat", "aliana",
                        "Antanta", "Puterile Centrale"]
      },
      "vectors": {
        "content_vector": [ ... ],
        "summary_vector": [ ... ]
      }
    }
  ]
}
```

3.3 Query understanding

Before searching, the system analyses the question in three successive stages (Figure 3.2). These stages have two major objectives: to avoid the costly processing of irrelevant questions and to produce filters that improve retrieval precision.

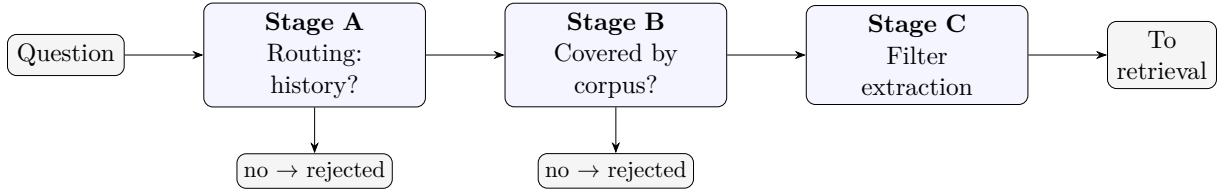


Figure 3.2: The query-understanding subsystem. Questions outside the domain or not covered by the corpus are stopped before retrieval.

3.3.1 Stage A — routing

The first stage decides whether the question belongs to the history domain. The decision is made by comparing the embedding of the question against two reference *anchors* — one for the “history” class, the other for “non-history” — using the cosine similarity of Equation (2.1). Questions outside the domain are rejected immediately, at minimal cost. This anchor-embedding approach is efficient, requiring only two anchors to be computed at initialization.

3.3.2 Stage B — the coverage check

The second stage checks whether the answer of the question is actually in the corpus. The decision combines three signals: a lexical score (a BM25 variant normalized by the sigmoid function, following Equation (2.2)), a semantic score (the maximum similarity to the chapter summaries), and the presence of *specific anchors* — entities, years, or concepts from the question that appear in the corpus metadata. Several decision rules combine these signals: for example, a high semantic score together with a specific anchor indicates coverage, while the absence of any strong signal leads to rejection. This also avoids hallucinations on topics outside the material.

3.3.3 Stage C — filter extraction

For accepted questions, the last stage extracts structured filters — years, locations, entities, periods, and key concepts — which will be used to narrow retrieval to the relevant passages. These filters are the link between question understanding and the retrieval mechanism.

3.4 Hybrid retrieval

Retrieval combines two complementary routes, whose results are merged and reranked (Figure 3.3).

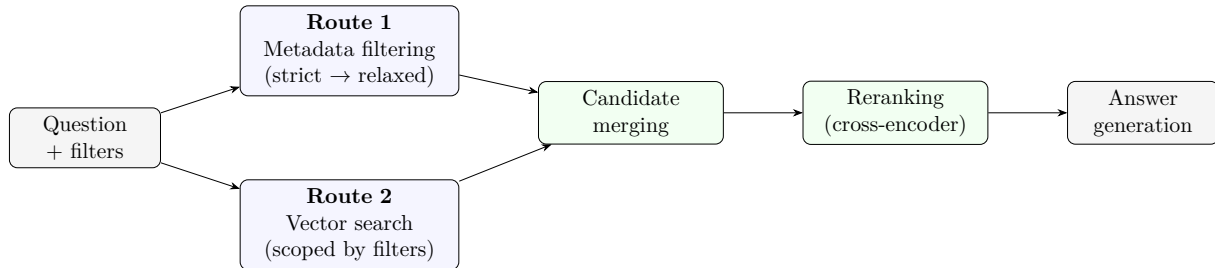


Figure 3.3: The hybrid retrieval mechanism: metadata filtering and vector search, followed by merging and reranking.

3.4.1 Route 1 — metadata filtering

The first route exploits the structured metadata extracted in Stage C. It applies a hard filter over the database, selecting the chapters that match exactly the entities, years, or locations in the question. The strategy has two steps: a *strict* match is tried first (all conditions satisfied); if it returns nothing, the system falls back to a *relaxed* match (a subset of the conditions). This route gives high precision when the question contains clear anchors.

3.4.2 Route 2 — vector search

The second route performs semantic search: the question is embedded, and chapters are retrieved by the similarity between the query vector and the summary vectors. The search is carried out preferentially through the database’s native vector search mechanism (`$vectorSearch` in MongoDB Atlas), with an automatic fallback to a local cosine search when it is not available. When Stage C has extracted precise filters (years or locations), the vector search is *scoped* to the corresponding documents, which makes it both more precise and cheaper.

3.4.3 Merging and reranking

The results of the two routes are merged with duplicates removed, and candidates from the *high-precision route* (Route 1, metadata filtering) are prioritized. The resulting set is then reranked by a dedicated *cross-encoder* (the Cohere Rerank service [1], model `rerank-v3.5`), which scores the relevance of each candidate to the question by processing the pair jointly, following the principle described in Chapter 2. Reranking brings to the

top the most suitable passages, which will form the context passed to the generative model.

3.5 Answer generation

The final stage conditions generation on the retrieved passages. The selected passages are assembled into a context with numbered sources, and the generative model (gpt-4o-mini) is instructed, through a *system prompt*, to follow three rules: write in Romanian in the simplest language possible, answer *only* from the context — inventing no dates, names, or years — and cite the source after each statement, in the form $[n]$. To reduce variability and the risk of hallucination, generation runs at a low temperature.

The system answers the student’s question using the retrieved context. If no relevant passage is available, the system states explicitly that it does not have enough information, rather than producing an unsupported answer. The $[n]$ citations allow every statement to be traced back to its source chapter, providing the transparency the educational setting requires.

3.6 Evaluation of Module 1

The module was evaluated on a built question set covering different lessons and levels of difficulty, and including both questions *covered* by the corpus and *uncovered* ones, in order to test behaviour at the domain boundary. The evaluation runs at three levels: routing, retrieval, and generation. The evaluation set comprised 220 questions (110 covered and 110 uncovered), distributed evenly across the lessons in the corpus.

3.6.1 Level 1 — routing

The first level evaluates the correctness of the query-understanding subsystem, separately for Stages A and B, through accuracy and F_1 score (Equation (2.5)). Table 3.1 reports the results. Domain classification (Stage A) is nearly perfect, and the coverage check (Stage B), a harder task, also reaches high values.

Table 3.1: Level 1 — evaluation of routing.

Component	Accuracy	F_1 score
Stage A (domain routing)	98.2%	98.9%
Stage B (coverage check)	87.2%	92.5%
Final routing (A + B)	85.5%	91.3%

3.6.2 Level 2 — retrieval

The second level measures retrieval quality through Hit@ k and MRR (Equation (2.4)), on the covered questions. Table 3.2 shows that the relevant document is retrieved in the top three results in more than 91% of cases, and an MRR close to 0.9 indicates that the relevant result usually appears in first position. The database holds roughly 270 documents to search over, from which only the five most relevant are finally selected.

Table 3.2: Level 2 — evaluation of retrieval.

Metric	Value
Hit@1	85.7%
Hit@3	91.4%
Hit@5	96.2%
MRR	0.886

3.6.3 Level 3 — generation

The third level evaluates the quality of the generated answers through the *LLM-as-a-Judge* method (Chapter 2). An independent judge model (Claude Sonnet 4.6), different from the one used to generate the answers (gpt-4o-mini), assigns scores on a scale from 1 to 5 along four dimensions: *faithfulness* (the answer invents no information), *relevance* (it answers the question directly), *accuracy* (the facts are correct against the ground truth), and *clarity* (the language suits a seventh-grade student). The results in Table 3.3 indicate high faithfulness to the context and clear answers, adapted to the target audience.

Table 3.3: Level 3 — evaluation of generation (1–5 scale).

Dimension	Mean score
Faithfulness	4.77
Relevance	4.31
Accuracy	4.08
Clarity	4.63
Overall mean	4.45

To isolate the contribution of the retrieval pipeline, the answers of the proposed system were compared with those of a *baseline*: the same generative model (gpt-4o-mini), queried through a plain API call, with no pipeline and no access to the lesson corpus — the answer produced solely from the model’s parametric knowledge. Both variants were evaluated on the same question set and under the same *LLM-as-a-Judge* protocol, the only difference being the presence or absence of grounding in the corpus (the “RAG system” column repeats the values from Table 3.3).

Table 3.4: Comparison between the baseline (LLM only) and the proposed RAG system (1–5 scale).

Dimension	LLM only	RAG system	Δ
Faithfulness	3.37	4.77	+1.40
Relevance	4.10	4.31	+0.21
Accuracy	2.77	4.08	+1.31
Clarity	4.50	4.63	+0.13
Overall mean	3.68	4.45	+0.77

The comparison shows clearly where retrieval contributes. The largest gains appear in *faithfulness* (+1.40) and *accuracy* (+1.31) — exactly the dimensions tied to factual grounding —, while *relevance* and *clarity* stay almost unchanged. The pattern is expected: a capable language model is already fluent and produces on-topic answers, but, lacking a source of truth, it frequently produces unfounded or incorrect statements. The baseline’s accuracy score (2.77) is especially problematic in an educational setting, since it shows that, without grounding, the system could pass incorrect information to the students. The value added by the RAG architecture therefore lies not in the quality of the language but in the correctness and grounding of the content — precisely the essential requirements for a teaching tool.

3.6.4 Qualitative analysis and examples

Beyond the aggregate metrics, this section examines two concrete cases — one successful and one failed — to show where the retrieval pipeline contributes and where its limitations appear. The system runs in Romanian, so the questions and answers are reproduced in the original.

Successful case. On well-anchored topical questions, hybrid retrieval succeeds consistently. Figure 3.4 illustrates this behaviour: the answer is phrased simply, and every statement carries a citation $[n]$ to the chapter it comes from. The question contains clear anchors (the entity “Primul Război Mondial”, the concept “asasinatul de la Sarajevo”), which Stage C extracts as structured filters. Route 1 selects the correct chapters directly, and reranking brings them to the top of the context. With all the necessary information in the retrieved passages, the generative model produces a complete answer, with no unfounded parametric knowledge.

Failure case. Behaviour degrades on narrow factual questions, especially those asking for an exact number. Figure 3.5 illustrates such a case, which reveals a real limitation but also a desirable property of the system. The *correct* behaviour is the refusal to invent: not finding the figure in the context, the model does not produce a plausible number but signals the missing information, thereby preserving high faithfulness to the sources.

Question: *Cum a izbucnit Primul Război Mondial?*

Generated answer: Primul Război Mondial a început după asasinatul de la Sarajevo, care a oferit Austro-Ungariei pretextul pentru a declara război Serbiei. Atacul a reprezentat o încercare de a elimina influența rusă din regiune și de a potoli revoltele slavilor din imperiu [1]. Declarația de război a avut loc pe 28 iulie 1914, iar conflictul s-a extins rapid datorită sistemelor de alianță existente [2].

Sources:

[1] Capitolul „Europa Alianțelor Politico-Militare”, lecția „Primul Război Mondial”.

[2] Capitolul „Desfășurarea Războiului”, lecția „Primul Război Mondial”.

Figure 3.4: Example of a generated answer, with citations pointing to the source chapter.

The limitation lies in retrieval: a question of the form “câți soldați au murit” offers no discriminating anchor (a year or a rare entity) that Stage C could turn into a filter, so selection falls back on vector search over summaries, optimized for topic-level retrieval. This correctly brings up the lesson on World War I, but it does not isolate the specific passage that would contain the figure — either because no such sentence exists in the indexed lessons, or because its summary similarity is too low to enter the top results. The most fragile dimension therefore remains factual accuracy on questions asking for an isolated detail, and the remedy lies in indexing granularity and in richer numeric metadata extracted at ingestion.

Question: *Câți soldați români au murit în Primul Război Mondial?*

Generated answer: „În contextul Primului Război Mondial, nu sunt menționate date specifice despre numărul soldaților români care au murit. Prin urmare, nu pot oferi o cifră exactă.”

Figure 3.5: Example of a controlled failure: on a narrow factual question, the system does not find the figure in the retrieved context and explicitly refuses to invent it, instead of producing an unfounded answer.

3.6.5 Discussion

The results confirm that the proposed architecture provides precise retrieval and answers faithful to the context, at low cost (fractions of a cent per query) and with a latency of a few seconds, values compatible with interactive use. The dimension with the greatest room for improvement is factual accuracy on questions asking for narrow details (an exact date, for example), where insufficiently specific context can lead to imprecision. This suggests, as a direction for refinement, tuning the granularity of the retrieved passages and the number of passages included in the context.

Chapter 4

Module 2: Detecting contradictions between sources

The second module tackles a problem specific to the study of history: identifying contradictions between several sources that treat the same topic. Unlike the first module, which assumes a coherent teaching corpus, this module starts from the assumption that sources may contain divergent claims and sets out to flag them automatically.

4.1 Motivation

History is a discipline in which interpretation plays a central role. The same series of events can be recounted differently depending on the source, either due to varying perspectives or because of errors and imprecision. For a student, meeting contradictory sources is both a difficulty and an opportunity to develop critical thinking. A tool that automatically identifies the points of divergence between sources can support this analysis by highlighting the claims that deserve close examination. The problem also illustrates a relevant case in natural language processing: verifying factual consistency between documents of comparable authority (two sources with a similar degree of trust). Without an external arbiter, this becomes a problem of logical compatibility — deciding whether two claims can be true at the same time.

4.2 Pipeline architecture

The module receives a set of sources on the same topic and produces a list of detected contradictions. Processing follows the stages in Figure 4.1: collecting the sources, aligning them through canonical questions, extracting typed claims enriched with entities, grouping by question and, finally, applying the detection mechanisms to the comparable pairs.

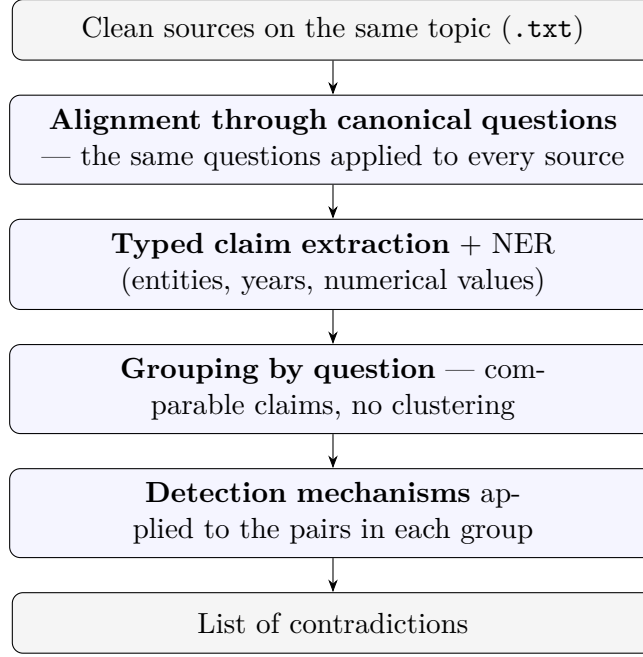


Figure 4.1: The contradiction-detection pipeline. Alignment on canonical questions guarantees that only claims about the same aspect are compared.

4.3 Aligning sources through canonical questions

The central challenge in detecting contradictions is *comparability*: a contradiction only makes sense between claims that refer to the same aspect. An approach based on automatically clustering claims by semantic similarity is fragile, since claims about different events but phrased similarly can receive nearly identical embeddings, producing mismatched comparisons and, consequently, false positives.

The solution adopted here removes this fragility by aligning the sources on a set of *canonical questions*. The same questions are applied to every source, and the answers obtained for a given question are comparable to one another by construction, because they refer explicitly to the same aspect. The hard problem of establishing comparability is thus solved at design time, not through approximate grouping.

4.4 Typed claim extraction and entity enrichment

For each (source, canonical question) pair we extract a *typed claim*, represented as a structure holding the answer itself, the text passage it comes from, and the aspect it targets. Each claim is then enriched through named entity recognition, using the Romanian model `ro_core_news_lg` from spaCy. From the claim we extract the entities mentioned, the years, and the numerical values. Entities are normalized to their base form (lemmatized) in order to neutralize Romanian inflection and allow the same entity to be matched across different sources. The typed representation — text, entities, years,

numbers — supports the detection mechanisms described below.

4.5 Detection mechanisms

For each group of claims associated with the same question, the claim pairs coming from different sources are evaluated by several complementary mechanisms, whose scores are then aggregated.

4.5.1 Structural conflict

The first mechanism targets explicit divergences between dates and numerical values. If two claims about the same aspect contain different years or figures (two different years for the same event, for example) beyond a given tolerance, the mechanism flags a structural conflict. This signal is very precise, since numerical divergences are rarely ambiguous, and it is particularly relevant in the historical domain, where dates and figures are central factual elements.

4.5.2 Natural language inference

The second mechanism applies a natural language inference (NLI) model, based on a multilingual variant of DeBERTaV3, to estimate the probability that one claim *contradicts* the other. Since the inference relation is directional, the pair is evaluated in both directions, keeping the maximum contradiction probability (Equation (2.3)). This mechanism captures contradictions expressed through language, without explicit numerical divergences — for example, claims that assign opposite roles to the same entities.

4.5.3 The entity anchor with fuzzy matching

To avoid comparing claims that, although associated with the same question, refer to distinct entities, we use an entity anchor. It checks the overlap of entities between the two claims through approximate (*fuzzy*) matching, tolerant of spelling variation. The matching also works *across languages* (Romanian–English), which allows correct anchoring when the sources use different forms of the same name. The role of this mechanism is to ensure that natural language inference fires only when the two claims refer to the same entity, reducing false positives.

4.5.4 The LLM judge

For pairs where the divergence is not structural but the signals remain ambiguous, an *LLM-as-a-Judge* is used as a fallback mechanism. It receives the pair of claims and

decides whether a real contradiction exists. The judge is called *per pair*, after the cheaper filters have been applied, which gives both robustness (a narrow context focused on a single comparison) and cost control, by limiting the number of calls.

4.5.5 Score aggregation

The final decision for a pair combines the mechanisms above. The structural and inference scores are aggregated by the maximum operator, $\text{score} = \max(\text{score}_{\text{struct}}, \text{score}_{\text{nli}})$, each with its own firing threshold. This rule works correctly precisely because the selection of comparable pairs was made upfront (Section 4.3) and gated by the entity anchor: without these filters, natural language inference would produce signals for unrelated pairs as well. Pairs whose score exceeds a verdict threshold are classified as contradictions.

4.6 Evaluation on an annotated corpus

4.6.1 Building the reference corpus

The evaluation uses manually verified topical corpora, in which the contradictions are *documented* in advance in a ground truth file. For each topic, the sources come with a metadata file describing their content and with a file listing the expected contradictions. This setup allows standard metrics to be reported by comparing automatically detected contradictions with documented ones. The corpora used cover topics in contemporary history — for example, Romania’s participation in the two World Wars. An important methodological point, identified during development, is the need for strict consistency between the reference file and the source actually evaluated: if the reference labels describe a different scenario from the one processed, the resulting metrics become meaningless. The alignment between ground truths and sources was therefore validated programmatically.

4.6.2 Results

Table 4.1 reports the results obtained on the test corpus, measured through recall, precision, and F_1 score (Equation (2.5)), against the documented contradictions. To account for the stochastic nature of the components based on language models — in particular the LLM-as-a-Judge — the evaluation was repeated ($n = 10$ runs), reporting the mean and standard deviation of each metric.

The values indicate balanced behaviour, with both metrics at a high level: a recall of 0.975 shows that nearly all documented contradictions are flagged, while a precision of 0.910 confirms that alignment on canonical questions (Section 4.3) and the entity anchor effectively limit false positives. Recall’s slight edge over precision reflects the rare cases in

Table 4.1: Contradiction detection on the annotated test corpus (mean $\pm$ standard deviation over $n = 10$ runs).

Metric	Value
Recall	0.975 ± 0.050
Precision	0.910 ± 0.045
F_1 score	0.940 ± 0.028

which a comparable pair is marked as divergent without corresponding to a documented contradiction; the F_1 score of 0.940 summarizes this well-balanced profile.

4.6.3 Qualitative analysis and examples

To illustrate the module's behaviour concretely, this section presents five real cases from the test corpora, giving the claims extracted from the sources, the typed elements, and the mechanism that produced the results. The sources are Romanian textbooks, so the passages and the extracted answers are reproduced in the original. The first two are structural contradictions (a date and a numerical value), the third is a purely semantic contradiction, the fourth is a correctly classified case of agreement that illustrates the precision threshold, and the last is a wrong detection, a classic false positive. Note that, for a measurable evaluation, the contradictions are *planted* in a controlled way in a synthetic source and confronted with the reference textbook. The synthetic source is a second textbook in which certain passages were altered so that contradictions could be extracted. This source uses different phrasing and structure from the reference textbook, thereby simulating the real case in which the authors and their styles differ.

Canonical question: <i>Când a intrat România în Primul Război Mondial?</i>
Textbook (seventh grade): „La 28 august 1916, România intra în război alături de Antanta pentru eliberarea românilor aflați sub stăpânirea austro-ungară și pentru întregirea statului.” <i>Extracted answer:</i> 28 august 1916
Synthetic source (planted error): „La 14 august 1916, România a intrat în război de partea Antantei, declarând război Austro-Ungariei.” <i>Extracted answer:</i> 14 august 1916
Mechanism: structural conflict — different days (28 vs 14 august 1916), confirmed also by natural language inference (contradiction probability ≈ 0.99).
Verdict: contradiction.

Figure 4.2: A structural date contradiction: the two sources give different days for the same event. Flagged simultaneously by the structural check and by NLI.

Canonical question: <i>Câte tancuri au folosit britanicii în bătălia de pe Somme?</i>
Textbook (seventh grade): „Anglo-francezii, revenind la războiul de mișcare, au declanșat ofensiva de pe Somme, în care britanicii au folosit pentru prima dată 47 de tancuri.” <i>Extracted answer:</i> 47
Synthetic source (planted error): „[...] ofensiva de pe Somme, în care britanicii au folosit pentru prima dată tancul: 32 de tancuri au participat la atacul inițial.” <i>Extracted answer:</i> 32
Mechanism: structural conflict — different figures (47 vs 32), confirmed by NLI (probability ≈ 0.98).
Verdict: contradiction.

Figure 4.3: A structural numerical contradiction: two sources give incompatible figures for the same aspect.

Canonical question: <i>Ce rol au avut crizele politico-militare în formarea alianțelor?</i> Source with the erroneous claim: „Crizele politico-militare au avut un rol minor și au contribuit la reducerea tensiunilor dintre marile puteri.” <i>Extracted answer:</i> rol minor; au redus tensiunile Reference textbook: „Crizele politico-militare — marocană (1905–1906, 1911), bosniacă (1908) și războaiele balcanice (1912–1913) — au mărit tensiunile dintre statele europene și au impus necesitatea constituirii unor alianțe militare.” <i>Extracted answer:</i> au mărit tensiunile; au impus constituirea alianțelor Mechanism: natural language inference (contradiction probability ≈ 0.94), confirmed by the LLM judge. There are no divergent numerical values — the structural check does not apply; the contradiction is semantic (an opposite role assigned to the same events). Verdict: contradiction.
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Figure 4.4: A semantic contradiction with no numerical divergence: the two sources assign the political and military crises of the early twentieth century (1905–1913) opposite roles. Only natural language inference and the LLM judge can handle this kind of case.

Canonical question: <i>În ce regiune a Europei s-au desfășurat Războaiele Balcanice?</i> Synthetic source: „În Balcani” Textbook: „În Peninsula Balcanică” Decision: <i>not</i> a contradiction (confidence ≈ 0.92). The two formulations name the same region; natural language inference stays below the contradiction threshold, and the pair is not reported.
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Figure 4.5: Example of correctly classified *agreement*: the same region (the Balkan Wars, 1912–1913) under two different names is not a contradiction. Such trap cases explain the precision threshold — the judge’s nondeterminism can mistakenly mark such a pair, generating the false positives (Figure 4.6) reflected in the precision of 0.91.

Canonical question: <i>Care a fost capitala de refugiu a României după ocuparea sudului țării în Primul Război Mondial?</i> Synthetic source: „Moldova”. Textbook: „Iași”. Result: the contradiction is detected in error. This is a classic false positive. The mechanism treats these as two different entities and therefore returns a contradiction verdict, without analysing the context with an LLM.

Figure 4.6: Example of a detection failure: although the synthetic source gives a valid answer (“Moldova”), an error is flagged, because the mechanism does not connect the two entities.

The first two cases (Figures 4.2 and 4.3) are structural contradictions, flagged with the highest confidence: a divergence in a date or a figure is rarely ambiguous, and on the full passages natural language inference confirms the same verdict. The semantic case (Figure 4.4) shows the limit of the structural check: with no divergent figure, only natural language inference and the LLM judge can recognize that one source assigns the political and military crises a role opposite to the other’s. The agreement in Figure 4.5 confirms that a small difference in wording — the same region under two names — is not treated as a contradiction but correctly kept as agreement. The false positive in Figure 4.6 shows the flip side of precision: when one source answers at the level of a region (“Moldova”) and the other at the level of a city (“Iași”), the mechanism treats the two names as distinct entities and flags a contradiction that does not exist, because recognizing that Iași lies in Moldova would require a contextual check that the structural filters do not perform.

A last, frequent case in which the pipeline produces errors is one where the incorrect information comes from a single source. The reason for the failure is structural: the module confronts only the claims answered by at least two sources, so an error present in a single source remains invisible, with nothing to compare it against.

4.6.4 Discussion

The results must be read within the limits of the evaluation corpus size: the metrics are reported on material from a narrow set of topics (the seventh-grade history syllabus) and were computed on the *in-scope* set, isolating the detection capability of the mechanisms from any coverage limitations of the canonical question set and of the sources. The F_1 score is also noticeably more stable (standard deviation 0.028) than recall and precision (0.045–0.050), which is explained by the two metrics partly compensating for each other from run to run, as the nondeterminism of the language-model judge occasionally moves a pair between categories. One methodological factor proved decisive: strict consistency between the reference file and the source actually processed, since a mismatch between the two produces meaningless metrics, no matter how good the detection mechanisms are.

Chapter 5

Conclusions

5.1 Summary of contributions

This thesis has presented a conversational system for history education, organized around two complementary modules that start from the same input — a question. The first module, a RAG-based assistant, gives answers grounded in the lessons and pitched at the level of a seventh-grade student. It integrates a complete data-ingestion pipeline (semantic chunking, metadata and entity extraction, hierarchical summarization, and embedding), a three-stage query-understanding subsystem, and a hybrid retrieval mechanism, followed by the generation of an answer with citations, faithful to the context. The second module addresses the detection of contradictions between historical sources by aligning them on canonical questions to secure claim comparability. The actual detection process then combines a structural check, natural language inference, an entity anchor, and an LLM judge.

The quantitative evaluation confirmed the effectiveness of both modules. For the RAG assistant, retrieval places the relevant document in the top three results in more than 91% of cases, and the generated answers obtain high faithfulness and clarity scores. For the contradiction-detection module, evaluation on an annotated corpus confirmed balanced behaviour, with faithful detection of the documented contradictions and a low rate of false-positive flags.

5.2 Limitations

The system does, however, have limitations, which narrow its use cases. The knowledge base is limited to the indexed lessons; questions outside this domain are rejected, which secures answer quality, but reduces coverage. Factual accuracy on questions asking for very narrow details remains the dimension with the greatest room for improvement. For the contradiction-detection module, the evaluation was carried out on small, con-

trolled corpora, so generalization to larger and more varied corpora remains to be confirmed. Both modules also depend on external services for embeddings, reranking, and generation, which introduces cost and availability considerations.

5.3 Future work

Several directions can extend this work. A first natural direction is enlarging the lesson corpus to cover the whole syllabus (for all lower-secondary grades), together with a re-evaluation of retrieval on this extended corpus. A second direction exploits the interaction between the two modules: the canonical questions used for contradiction detection can be reused as assessment items (quiz-style) in the pedagogical module, giving students exercises generated from the base material. A third direction suggests refining generation by tuning context granularity to improve accuracy on narrow questions. Finally, an extended validation with history teachers would strengthen the qualitative evaluation of the answers and of the flagged contradictions.

5.4 Personal remarks

Developing this system made clear that the value of an application built on language models comes not only from the generative model but, in equal measure, from the careful design of the components around it — how the data is prepared, how queries are understood, how information is retrieved, and how the answer is controlled. For contradiction detection, the most important lesson was that a hard problem — establishing comparability — can be solved more effectively by an early design decision than by added complexity later. Beyond this particular application, these observations offer useful guidance for building similar systems, factually grounded and adapted to a specific group of people.

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